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Neuro-Symbolic Reasoning Engines on Heterogeneous Scientific Knowledge Graphs

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ABSTRACT

Pure deep learning architectures struggle with multi-hop logical deductions on scientific knowledge graphs, often generating unverifiable extrapolations. We introduce a neuro-symbolic reasoning engine that unifies differentiable graph embeddings with first-order predicate logic solvers. Our approach translates knowledge subgraphs into probabilistic horn clauses, enforcing ontological consistency constraints during inference. On scientific fact verification datasets, our engine achieves 91.4% precision, outperforming standard relational graph convolutional networks by 14.2%. Nevertheless, the combinatorial complexity of the symbolic solver bounds the maximum clause depth to four hops, creating an expressivity bottleneck.

Keywords— Artificial Intelligence, Neuro-Symbolic, Knowledge Graphs, Logical Inference

INTRODUCTION

Scientific discovery demands automated reasoning that is both statistically flexible and logically rigorous. While neural graph encoders capture latent entity affinities, they lack axiomatic guarantees. We propose a hybrid framework combining neural node embeddings with discrete symbolic clause deduction.

METHODOLOGY

The engine constructs differentiable logic trees over relational graph edges. Neural encoders propose candidate logical paths, which the symbolic engine filters against domain axioms defined in Web Ontology Language (OWL).

RESULTS

Testing across biomedical and materials science knowledge graphs demonstrated state-of-the-art link prediction and theorem verification. The logical consistency constraint eliminated 96% of nonsensical edge predictions produced by pure neural baselines.

LIMITATIONS

The primary limitation is clause expansion depth. When reasoning paths require five or more relational hops, the branch-and-bound symbolic solver encounters significant memory pressure and runtime degradation.

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